

Jun (Jennifer) Luu

EDUCATION

University of Pennsylvania
Master of Urban Spatial Analytics
Weitzman School of Design
May 2026

Virginia Tech
B.S. in Construction Engineering and Management
College of Engineering
May 2022

TECHNICAL SKILLS

PYTHON	pandas, GeoPandas, scikit-learn, SciPy, NumPy, Rasterio, Shapely
R	tidyverse, tidycensus, sf, leaflet, ggplot2
MACHINE LEARNING & AI	Self-Organizing Maps, TF-IDF, sentence embeddings, Random Forest, k-means, Anthropic API, synthetic data generation
SPATIAL ANALYSIS	ArcGIS Pro, QGIS, PostGIS, Google Earth Engine, Mapbox, CARTO, Moran's I / LISA, spatial cross-validation
DATA & VISUALIZATION	SQL, Jupyter, Quarto, Git / GitHub, Seaborn, Plotly, HTML / CSS / JS, Excel, Looker, Adobe InDesign

RESEARCH

Mapping New Vocabularies of Neighborhood Change: An Inductive Text Analysis of Real Estate Listings

2025 – 2026

Pls: Prof. Elizabeth Delmelle (University of Pennsylvania) & Prof. Isabelle Nilsson (UNC Charlotte)

Presented at AAG Annual Meeting, San Francisco, March 2026

- Developed a generalizable inductive text analysis pipeline applying **TF-IDF vectorization** to MLS listing remarks across Mecklenburg County, NC to identify latent vocabulary patterns signaling neighborhood change across census tract-periods.
- Implemented a **Self-Organizing Map (SOM)** trained on listing language to cluster neighborhoods by linguistic profile, enabling unsupervised detection of gentrification stages without imposing predetermined typologies.
- Analyzed temporal sequences of SOM cluster membership across tract-periods to construct a **transition matrix** of neighborhood trajectories, tracing how individual tracts move through discursive market segments over time.

Synthetic MLS Remark Generation via Batch API

2025

Support pipeline for thesis research

- Designed and built a **batch API pipeline** using the Anthropic API to generate synthetic real estate listing remarks conditioned on neighborhood type, price tier, and time period, augmenting sparse tract-period cells in the Mecklenburg County dataset.
- Generated remarks are flagged with an **is_synthetic field** for downstream validation and sensitivity analysis, enabling direct comparison of model behavior with and without synthetic augmentation.
- Pipeline processes tract-period pairs with fewer than a minimum observation threshold, matching the style, vocabulary, and temporal register of real MLS remarks using few-shot prompting from real data samples.

Predicting Eviction Risk in Philadelphia: A Spatiotemporal Modeling Approach

2025

Collaborative with Jed Chew, Mark Deng, and Mohammad AlAbbas · MUSA 5080, University of Pennsylvania

- Cleaned and joined Eviction Lab filings (2020–2025), Philadelphia crime records, 311 service requests, ACS census variables, and real estate tax delinquency data into a unified monthly panel dataset across all Philadelphia census tracts.
- Engineered features including a **tract vulnerability index**, distance-to-hotspot measures, and temporal lag variables (1, 3, 6, and 12 months) to capture spatial and temporal structure in eviction filing patterns.
- Applied **Moran's I** and **LISA** spatial autocorrelation diagnostics to validate spatial structure, then implemented a **Poisson regression model** with spatial and temporal lags to forecast eviction filings by census tract.

RESEARCH POSITIONS

Research Specialist II

2026 – Present

EVICTION LAB — Princeton University, Princeton, NJ

Data and Spatial Analytics Fellow

2026

ONE ARCHITECTURE & URBANISM (ONE) — New York, NY

- Built a **relational database** cataloging the firm's mixed-format project archive (.jpg, .png, .ai, .pdf, and other file types), turning a previously unsearchable store of work throughout Box and Drive into a queryable resource.
- Developed a **crawler** that indexed the firm's full file holdings and populated the database automatically, so the catalog stayed current without manual entry.

Research Analyst

2026

PENN INSTITUTE FOR URBAN RESEARCH — University of Pennsylvania, Dr. Eugenia Birch

- Wrote foundational research memo defining scope, analytical framework, and survey strategy for a multi-year study on Penn's role as an anchor institution, informing research on a **\$37.7B annual regional economic footprint**.
- Developed and mapped a spatial inventory of Penn's full regional footprint using **ArcGIS Pro**, **R**, and **Adobe InDesign**, producing deliverable-quality maps of West Philadelphia and the broader metro region.
- Designed survey frameworks across major stakeholder populations: university deans, regional business owners, and health system patients and professionals.

Geospatial Research Assistant

2025 – 2026

CENTER FOR AIR TRANSPORTATION RESILIENCE (CATRES) — NASA University Leadership Initiative, Dr. Megan Ryerson (Penn) & Dr. Mark Hansen (UC Berkeley)

- Constructed an end-to-end data pipeline processing **25.4 million U.S. flight records** across 45 months (BTS + FAA) with fully reproducible, modular notebooks tracked in Git.
- Developed a temporal clustering algorithm isolating **64,815 diversions across 377 airports**, distinguishing system-wide cascade events from regionally contained disruptions.
- Built an interactive **Plotly / Panel dashboard** and Quarto documentation site translating findings into actionable recommendations for airline operations, airport capacity planning, and FAA policy.
- Identified that American Airlines diverted **29% more frequently than Delta** despite comparable network size, with implications for hub scheduling and operational resilience policy.

PROFESSIONAL EXPERIENCE

Assistant Project Manager

2024 – 2025

FAIRSTEAD — NYCHA Reid Apartments PACT/RAD Conversion, Brooklyn, NY

- Coordinated occupied renovation of **16,000+ NYCHA Section 8 housing units** under federal housing preservation compliance, managing workflows across relocation coordinators, construction teams, property management, the NYC housing authority, and resident liaisons.
- Designed and implemented a **unit-level progress tracking dashboard** monitoring renovation status across all units, enabling real-time decision-making on construction sequencing and tenant relocation to minimize disruptions.
- Researched Section 8 regulatory compliance and housing preservation frameworks to inform phased construction strategy under the PACT/RAD conversion program.

Field Engineer

2023 – 2024

HENSEL PHELPS — Harbor-UCLA Medical Center, Los Angeles, CA

- Developed logistics and scheduling strategies for complex infrastructure systems on a major healthcare campus serving majority low-income and uninsured patients in LA County.
- Audited cost estimates and identified discrepancies, **preventing \$250,000 in unaccounted material costs**.
- Collaborated with cross-functional teams to resolve technical and operational challenges across multiple active subcontractors.

Assistant Engineer

2021 – 2023

TURNER CONSTRUCTION COMPANY — Advocate Illinois Masonic Medical Center, Chicago, IL

- Coordinated seven subcontractors managing specialized systems on a major occupied hospital expansion with complex phasing to maintain full operations throughout construction.
- Analyzed contract documentation and budget data to identify cost and schedule risks, **value-engineering over \$100,000 in savings** on door upgrade specifications.
- Implemented a resource allocation tracking system monitoring personnel and material staging daily, optimizing site logistics for parking, equipment access, and workflow efficiency.

PRESENTATIONS

2026 *Mapping New Vocabularies of Neighborhood Change: An Inductive Text Analysis of Real Estate Listings*

American Association of Geographers Annual Meeting — San Francisco, CA, March 2026

LEADERSHIP & COMMUNITY

Women Build Team Lead

2023

HABITAT FOR HUMANITY — CHICAGO, IL

Led a team of 10 women to construct permanent affordable homes. Coordinated logistics, training, and team dynamics to ensure project completion and positive community impact.

Mixed-Use Team Competition Coach

2023 – 2024

USC CMAA — LOS ANGELES, CA

Created and taught lesson plans on construction project management, cost estimating, scheduling, and superintending. Prepared team for collegiate competition through technical training and mentorship.